UĞURKAN YİĞİT ÖZTÜRK

Software Engineering Student | AI & Computer Vision | Backend Development

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PROFILE

Software Engineering student focused on artificial intelligence, computer vision and backend development. Hands-on experience building practical AI prototypes involving YOLO-based vehicle analysis, OCR and document understanding, local LLMs and Retrieval-Augmented Generation (RAG). Experienced across complete workflows including data processing, model integration, REST APIs, databases, testing and real-world inference.

WORK EXPERIENCE

Software Engineering Intern — SANKO Holding / SANShine | Gaziantep, Türkiye **Summer 2026**

- Developed and tested AI-focused software prototypes across computer vision, OCR, local LLM/RAG and backend development.
- Built REST API, database and model-inference components using Python, FastAPI, PostgreSQL and locally hosted AI models.
- Worked on vehicle analysis, document understanding and intelligent website monitoring; performed debugging, performance checks and failure-case analysis.

EDUCATION

B.Sc. in Software Engineering — Ankara University | Ankara, Türkiye **2023 - Present**

Relevant coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks

SELECTED PROJECTS

Vehicle Vision System | YOLOv8 · Computer Vision · Python **Aug - Sep 2026**

- Built a fixed-camera pipeline for vehicle detection and 8-class vehicle-type classification covering automobile, SUV, pickup, truck, tractor-trailer, motorcycle, bus and panel van/minivan.
- Trained YOLOv8n-cls on 13,528 training images; achieved 96.9% Top-1 validation accuracy and evaluated the system on held-out traffic videos with false-positive and temporal prediction analysis.
- Separated vehicle detection, vehicle-type classification and wheel-detection tasks to keep the pipeline modular and support future axle, color and brand/logo analysis.

RAG-Based AI Assistant | RAG · LLM · FastAPI · PostgreSQL **Jul - Aug 2026**

- Built a local document-grounded assistant with PDF ingestion, text chunking, embedding generation, PostgreSQL vector storage, semantic retrieval and LLM-based response generation.
- Integrated Gemma/Llama-family models through Ollama and designed the architecture as a reusable template for multiple assistant profiles.
- Used nomic-embed-text embeddings and PostgreSQL-based retrieval, and tested chunking/retrieval settings to improve answer grounding.

Identity & Document OCR System | OCR · PaddleOCR · Computer Vision **Aug 2026**

- Developed local OCR workflows for Turkish ID cards, driver's licenses and personnel cards with document alignment, image preprocessing, field extraction, validation and structured JSON output.
- Implemented document-specific profiles and camera-based testing; optimized retries and post-processing for practical OCR performance.
- Applied field-specific validation such as Turkish identity-number checks and separate normalization rules for general text and location fields.

Website Monitor AI | Python · PostgreSQL · AI Analysis **Aug 2026**

- Developed backend components for scheduled webpage monitoring, snapshot comparison and topic-based change analysis, with PostgreSQL models for monitors, snapshots and detected changes.
- Tested extraction, similarity comparison, repository/database operations and AI-assisted topic analysis before notification-layer integration.

Zero Trust Network Access System | Keycloak · OAuth2 · OIDC · JWT · RBAC **Nov 2025**

- Implemented centralized authentication and granular authorization based on Zero Trust principles using Keycloak and standard identity protocols.
- Used OAuth2, OpenID Connect, JWT and RBAC to structure secure identity, authentication and role-based access flows.

TECHNICAL SKILLS

Programming / AI	Python, Java, YOLOv8, Computer Vision, OCR, PaddleOCR, NLP, LLM, RAG, Embeddings
Backend / Data	FastAPI, REST APIs, PostgreSQL, pgvector, Firebase, React
Security	Keycloak, OAuth2, OpenID Connect, JWT, RBAC
Tools	Git, Docker, Ollama

LANGUAGES & TRAINING

Turkish: Native **English:** Advanced

Training: Python ile Yapay Zeka ve Veri Bilimi / Python Programming - Atıl Samancıoğlu (Udemy); Artificial Intelligence Trainings - Prof. Dr. Murat Karakaya